

Sustainability Metrics and Reporting Framework

Chemical Industry Business Responsible Sustainability Reporting Analysis

Tata Chemicals Limited
Gujarat Fluorochemicals Limited
Navin Fluorine International Limited

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This report examines the Business Responsibility and Sustainability Reports (BRSR) of the chemical industry. Considering the company performance by market capitalization ranking and availability of reports, choose Tata Chemicals Limited (TCL), Gujarat Fluorochemicals Limited (GFL), and Navin Fluorine International Limited (NFIL) for three financial years. TCL operates as a top 1000 listed corporation marked by the Securities and Exchange Board of India, as a diversified, multinational with reporting across geographical locations. GFL is a Gujarat based fluorochemical manufacturer reporting on a standalone basis, while NFIL is a leading fluorochemical and Contract Development and Manufacturing Organisation among international market players with high export background.

For the analysis, parts of two sections and principles related to the adherence to national guidelines for responsible business conduct, environment sustainability and human rights and compliance systems and other indicators as metrics are considered. Among the three, TCL stands at the top of the reporting in terms of scale and data depth, while NFIL follows the alignment towards global sustainability certifications. GFL complies with regulatory disclosure norms but provides relatively lean narrative depth.

Our group divided the workload indicator by indicator and compared both the BRSR Reports and the respective XBRLs. In-depth analysis was conducted individually followed by group discussion to strengthen our understanding and develop genuine and correct conclusions and observations.

Company Profile

TATA CHEMICALS LIMITED



The main company is Tata Chemicals Limited (TCL) and as of December 2024, it ranks 283 based on the average market capitalization (₹27547.23 crores). It is a sustainable chemistry solutions company and produces distinguishable science-driven products. The company was incorporated in the year 1939 and is headquartered in Mumbai. Tata Chemicals operates across four continents and manages 15 manufacturing facilities. A few of its products are: soda ash, sodium bicarbonate, salt, agrochemicals and seeds. It serves the Indian subcontinent, America, South East Asia, Middle East, Africa, European Union and the United Kingdom (Tata Chemicals Limited, Integrated Annual Report : FY 22-23).

Gujarat Fluorochemicals Limited (GFL) is an Indian Chemicals Company that has been selected as one of the two peer companies for the assessment. GFL has 30 years of expertise in Fluorine Chemistry. They hold domain expertise in Fluoropolymers, Fluorospecialties, Refrigerants, and Chemicals. Apart from being one of the world's largest producers of fluoropolymers, GFL is also well-established and is a reliable partner to markets across Europe and the USA. They have established their presence globally in 16 different countries. The revenue performance has increased from ₹4,281crores to ₹4,737 crores in FY 2023-24.

GUJARAT FLURO CHEMICALS LIMITED



NAVIN FLUORINE INTERNATIONAL LIMITED



Navin Fluorine International Limited (NFIL), established in 1967, is one of India's leading specialty-fluorochemicals manufacturing companies under the Padmanabh Mafatlal Group. The company produces a range of fluorine-based products including refrigerants (sold under the brand name "Mafron"), inorganic fluorides, specially organofluorines. NFIL operates multiple manufacturing complexes across India (Surat & Dahej in Gujarat; Dewas in Madhya Pradesh) and runs a dedicated R&D center (the Navin Research Innovation Center, NRIC) to support development of novel fluorochemicals. For the fiscal year ended March 31, 2025, NFIL reported consolidated revenue from operations of ₹2,349 crores.

Section B: Management and Process Disclosures

Essential Indicator 10: Details of Review of NGRBCs by the Company

The disclosures provided by the Tata Chemicals Limited for FY2022-23, FY2023-24, and FY2024-25 under the Essential Indicator 10, have remained consistent for all the three years including the frequency of reviews and the authority responsible. However, the reviews provided for FY2023-24 and FY2024-25, are more prominent and aligns to BRSR guidelines.

Year on Year trends of Tata Chemicals Limited

The disclosures provided in FY 2022-23 does not provide a clear review of the NGBRC principles that are being accounted by the Tata Chemicals Limited. However, for FY 2023-24 and FY 2024-25, reviews appropriately and uniformly on all the NGBRC principles that are being implemented and followed by the company but no common standard is followed i.e. for FY 2023-24 another way of representation is used, while for FY 2024-25, another way is used.

The company has consistently mentioned about the frequency of review being done periodically or a on a need basis. These reviews are being undertaken at the senior leadership and Board level across the three years, that involves the Managing Director and CEO, to ensure top-level accountability.

10. Details of Review of NGRBCs by the Company:										10. Details of Review of NGRBCs by the Company:									
Subject for Review	Indicate whether review was undertaken by Director / Committee of the Board/ Any other Committee									Subject for Review	Indicate whether review was undertaken by Director / Committee of the Board/ Any other Committee								
	P1	P2	P3	P4	P5	P6	P7	P8	P9		P1	P2	P3	P4	P5	P6	P7	P8	P9
Performance against above policies and follow-up action										Performance against above policies and follow-up action	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Compliance with statutory requirements of relevance to the principles, and, rectification of any non-compliances										Compliance with statutory requirements of relevance to the principles, and, rectification of any non-compliances	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
As a practice, Business Responsibility policies of the Company are reviewed periodically or on a need basis by Senior Leadership Team including Managing Director & Chief Executive Officer. During this assessment, the efficacy of the policies is reviewed and necessary changes to policies and procedures are implemented.										As a practice, Business Responsibility policies of the Company are reviewed periodically or on a need basis by Senior Leadership Team including Managing Director & Chief Executive Officer. During this assessment, the efficacy of the policies is reviewed and necessary changes to policies and procedures are implemented.									
The Company is in compliance with the existing regulations as applicable and a Statutory Compliance Certificate on applicable laws is provided by the Managing Director & Chief Executive Officer / Chief Financial Officer / Chief General Counsel & Company Secretary to the Board of Directors.										The Company is in compliance with the existing regulations as applicable and a Statutory Compliance Certificate on applicable laws is provided by the Managing Director & Chief Executive Officer / Chief Financial Officer / Chief General Counsel & Company Secretary to the Board of Directors.									

FY 2022-23

FY 2023-24

FY 2024-25

Peer Comparison of Tata Chemicals Limited with Gujarat Fluorochemicals Limited and Navin Fluorine International Limited

10. Details of Review of NGRBCs by the Company:										10. Details of Review of NGRBCs by the Company:									
Subject for Review	Indicate whether review was undertaken by Director / Committee of the Board/ Any other Committee									Subject for Review	Indicate whether review was undertaken by Director / Committee of the Board/ Any other Committee								
	P1	P2	P3	P4	P5	P6	P7	P8	P9		P1	P2	P3	P4	P5	P6	P7	P8	P9
Performance against above policies and follow-up action										Performance against above policies and follow-up action	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Compliance with statutory requirements of relevance to the principles, and, rectification of any non-compliances										Compliance with statutory requirements of relevance to the principles, and, rectification of any non-compliances	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
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TATA CHEMICALS LIMITED, FY 2024-25

NFIL, FY 2024-25

GFL, FY 2024-25

TCL and GFL has used a similar way of representation but NFIL has used different way of representation of the indicator 10 in their FY 2024-25, still all three of the reports have not used the standard format.

Section C: Principle Wise Performance Disclosure

Principle 2: Businesses should provide goods and services in a manner that is sustainable and safe.

Essential Indicator 2

- Does the entity have procedures in place for sustainable sourcing? (Yes/No)
- If yes, what percentage of inputs were sourced sustainably?

Overall, there has been a high connectivity of information in our main company and its peers. They all have reported Yes to sourcing sustainably, followed by the percentage of inputs.
For Tata Chemicals Limited, the difficulty came in connecting the information in FY 23-24 as they had only included the materials sourced sustainably in its Indian operations but the company operates across four continents and manages about 15 manufacturing facilities.

Parameter	Remarks
Tata Chemicals Limited	A very minute difference in the FY 22-23, BRSR Report: 19% of inputs were sourced sustainably, while XML : 18.9%. In FY 23-24, TCL said that they source 28.85% sustainably, but in notes they mentioned that it is only for TCL India. In FY 24-25, they sourced 24.99% of their sources sustainably. As FY 23-24 only included the source from TCL India, it becomes a little hard to connect the information and compare if they have increased their sustainable sourcing.
Gujarat Fluorochemicals Limited and Navin Fluorine International Limited	FY 24-25: Both NFIL and GFL gave more description than just yes to 'Does the entity have procedures in place for sustainable sourcing'. TCL is a step ahead and has disclosed that they were reasonably assured for the BRSR Core Indicators under the category 'Assurance sub type for the entity have procedures in place for sustainable sourcing and percentage of inputs were sourced sustainably'.

2. Does the entity have procedures in place for sustainable sourcing? (Yes/No) If yes, what percentage of inputs were sourced sustainably?	
Entity has procedures (Yes / No)	Yes
Percentage of inputs	28.85*
*For TCL India	

Figure : BRSR Tata Chemicals Limited - FY 23-24

2.	a.	Does the entity have procedures in place for sustainable sourcing? (Yes/No)
		Yes, the Company has developed a supplier sustainability code and has established process for vendor selection. This includes various principles and guidelines such as Safety, Health and Environment Policy, Legal Compliance, adherence to TCoC, ISO Certification, etc. The Company has started carrying out a Sustainability assessment of its key suppliers and communicates areas of further improvements to reinforce sustainability principles.
	b.	If yes, what percentage of inputs were sourced sustainably?
		19%

<in-capmt:DetailsOfImprovementsInEnvironmentalAndSocialImpactsDueToCapex contextRef="DCMain">Projects for Pollution Control, Safety for Employee & Community, Climate Change, Circularity i.e. harnessing solar energy into electricity, capex for energy storage materials research, human health and well-being and sustainable chemicals.</in-capmt:DetailsOfImprovementsInEnvironmentalAndSocialImpactsDueToCapex>
<in-capmt:DoesTheEntityHaveProceduresInPlaceForSustainableSourcing contextRef="DCMain">true</in-capmt:DoesTheEntityHaveProceduresInPlaceForSustainableSourcing>
<in-capmt:PercentageOfInputsWereSourcedSustainably contextRef="DCMain" decimal="INF" unitRef="pure">0.189</in-capmt:PercentageOfInputsWereSourcedSustainably>
<in-capmt:DescribeTheProcessesInPlaceToSafelyRecycleYourProductsForReusingRecyclingAndDisposingAtTheEndOfLifeForPlasticsIncludingPackagingExplanatoryTextBlock contextRef="DCMain">Damaged material is brought back to the plant, and reprocessed as per SOP through contracting process. For exported material, customers are required to safely dispose off the product as per local regulations.</in-

Figure : BRSR & XBRL of Tata Chemicals Limited - FY 22-23

Principle 5: Businesses should respect and promote human rights.

Essential Indicator 10: Details of any corrective actions taken or underway to address significant risks / concerns arising from the assessments at Question 9.

TATA CHEMICALS LIMITED (TCL)			
Complain type	Filed during current FY	Filed last FY	Description provided by the entity
Sexual Harassment	1	1	The Company internally monitors compliance with all relevant laws and policies pertaining to these issues at 100% of its plants, offices & other establishments. There have been no observations by local statutory / third parties in FY 2024-25.
Discrimination at Workplace	3	6	
Child Labour	0	0	
Forced Labour/ Involuntary Labour	0	0	
Wages	0	0	
Others	8	18	

GUJARAT FLUROCHEMICALS LIMITED (GFL)			
Complain type	Filed during current FY	Filed last FY	Description provided by the entity
Sexual Harassment	0	0	100 % of plants and offices that were assessed (by the entity or statutory authorities or third parties)
Discrimination at Workplace	0	0	
Child Labour	0	0	
Forced Labour/ Involuntary Labour	0	0	
Wages	0	0	
Other human rights-related issues	1	0	

December 29, 2024: New article reported four workers died after inhaling toxic fumes at the Dahej/CMS plant (national news: NDTV, Indian Express, Economic Times, Business Standard)

NAVIN FLUORINE INTERNATIONAL LIMITED (NFIL)			
Complain type	Filed during current FY	Filed last FY	Description provided by the entity
Sexual Harassment	0	0	100 % and Nil plants and offices that were assessed (by entity or statutory authorities or third parties)
Discrimination at Workplace	0	0	
Child Labour	0	0	
Forced Labour/ Involuntary Labour	0	0	

Principle 6: Businesses should respect and make efforts to protect and restore the environment.

Essential Indicator 1: Total Energy Consumption (in Joules or multiples) and Energy Intensity.

There have been major changes in the taxonomy from FY 22-23 to FY 23-24 as they have moved the leadership indicators to the essential indicators for the Total Energy Consumption (in Joules or multiples) and Energy Intensity disclosure.

Every year Tata Chemicals Limited has updated their reporting boundaries and restated their values to enable “comparability of information”, but it is doing the opposite. We were unable to actually assess if they are getting environmentally friendlier or not. The results are inconsistent and incomparable the first time we look at it. We would have to put extra efforts, such as checking if the restated values are actually true or understated. The good thing is that they have made us aware about the change in the footnotes.

Year on Year trends of Tata Chemicals Limited

XBRL and BRSR reports for TCL for FY 22-23 are completely off the charts and the data in XBRL file is incorrect as the XBRL file uses unitRef = "pure" which is essentially used for dimensionless indicators such as ratios, percentages, boolean, etc. The correct way would have been to assign it a SI unit such as joules or its multiples.

Independent Evaluation

FY 24-25: All the companies have stated Yes and the names of the assurance agencies. All have reasonable assurance in 'Assurance sub type for details of total energy consumption in joules or multiples and energy intensity' .
But, TCL is a step ahead and has All in 'Type of assurance for principle 6 essential indicators', while the other two have 'Partial'.

Peer Comparison of Tata Chemicals Limited with Gujarat Fluorochemicals Limited and Navin Fluorine International Limited

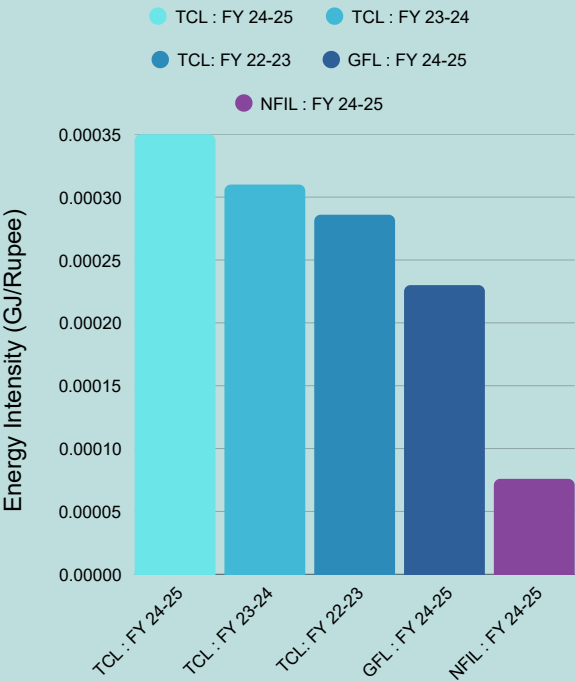
From Renewable Sources Parameter: All values were correct for FY 24-25 across the BRSR Report and the XBRL files.
Suggestion: TCL should have put 0 instead of - in their BRSR Report.

From Non-Renewable Sources Parameter: For NFIL, Total energy consumed from non-renewable sources differs by an extremely small amount, it is 1726617.12 instead of 1726617.13 (BRSR Report). NFIL - 1596686 instead of 1596685.96 (BRSR).
Suggestion: TCL should have put 0 instead of - in their BRSR Report.

Energy Intensity

FY 24-25: NFIL has the lowest energy intensity i.e. it consumes the least energy per rupee of turnover. Then comes GFL at 0.00023, followed by TCL performing the worst at 0.00035. This is because a lower energy intensity level indicates that the company has lower operating costs, a competitive advantage to energy price fluctuations, resource efficiency and a lesser environmental impact.

TCL across FY 22-23, 23-24 and 24-25: The energy intensity of the company has been on a rise from 0.00027 to 0.00031 to 0.00035 GJ/Rupee in FY 24-25. The company should lower its energy intensity to harness the benefits mentioned above.
NOTE : In FY 22-23, the company used the unit TJ while in the subsequent years, it switched to GJ to measure energy.



Essential Indicator 3: Water withdrawal by source (in kilolitres) and Water Intensity.

During the analysis of the BRSR reports submitted by the Tata Chemicals Limited for FY 2022-23, FY 2023-24, and FY 2024-25, and by Gujarat Fluorochemicals Limited and Navin Fluorine International Limited for 2024-25, the disclosures provided were found to have few mismatches and misleading entries. The reports did align to the required formats of BRSR. All of the provided values associated with the amount of water withdrawal are provided in Kilolitres (KI), that aligns with the given BRSR criteria.

Year on Year trends of Tata Chemicals Limited

For the part of Water intensity per rupee of turnover, the XBRL and PDF BRSR reports for all three years were found to be mismatched or misleading. Whereas the XBRL has shown the original water intensity per rupee (till 8-9 decimal points) but the values in PDF have been rounded off to only 5 decimal points, which is leading to misinterpretation. This case is for financial years 2023-24 and 2024-25, for 2022-23 it was the same but it was rounded off to 5th decimal place. All other parameters were also found to be similar in both the XBRL and PDF of BRSR report.

Independent Evaluation

FY 24-25: All the companies have stated Yes and the names of the assurance agencies. All have reasonable assurance for Water Withdrawal.

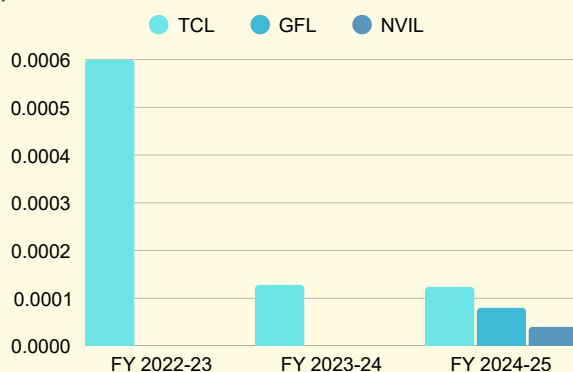
But, TCL is a step ahead and has All in 'Type of assurance for principle 6 essential indicators', while the other two have 'Partial'.

Water Intensity

For FY 2024-25, NFIL has the lowest water intensity i.e. 0.0000447557 which means, it withdraws the least water per rupee of turnover.

Then comes GFL at 0.0000808268, followed by TCL at 0.0001242863. A lower water withdrawal intensity level indicates that the company has lower environmental effects on groundwater depletion, an advantage of being called a more sustainable and resource efficient company.

TCL across FY 22-23, 23-24 and 24-25: The Water withdrawal intensity of the company has been on a decline from 0.0006 (2022-23) to 0.000128 (2023-24) to 0.000124 kI/Rupee in FY 24-25.



Water intensity per rupee of turnover (for FY 2022-23, 2023-24, 2024-25).

Peer Comparison of Tata Chemicals Limited with Gujarat Fluorochemicals Limited and Navin Fluorine International Limited

In the peer comparison, there also the mismatch was found in the representation of the water intensity per rupee of turnover.

TCL has rounded off the original values till 5 decimal points, which is there in the XBRL as 8-9 decimals.

The similar issue is with the Gujarat Fluorochemicals Limited, they have also rounded off the water intensity values to 6 decimal points for both 2024-25 and 2023-24 reports.

FY 2023-24		FY 2024-25	
1758	WaterIntensityPerRupeeOfTurnover	01-04-2023 To 31-03-2024	DCYMain
1759	WaterIntensityPerRupeeOfTurnover	01-04-2022 To 31-03-2023	DPYMain
Water intensity per rupee of turnover (Total Water consumption / Revenue from operations)		0.00013	0.00011
Water intensity per rupee of turnover adjusted for Purchasing Power Parity (PPP) (Total water consumption / Revenue from operations adjusted for PPP)		0.0029	0.00242

Misrepresentation of values by TCL by rounding off the original values of water intensity for FY 2023-24

Navin Fluorine International Limited has shown the values of the same parameter in PDF in exponential form but in the XBRL it is in regular decimal form.

One other thing is that Tata Chemicals Limited and Navin Fluorine International Limited have mentioned the 0 values as hyphen (-) in their pdf reports, but in the report of Gujarat Fluorochemicals, 0 has been mentioned as 0 in both the XBRL and PDF.

All other values were found to be correctly represented for FY 24-25 across the BRSR Report and the XBRL files by all the three companies.

Water intensity per rupee of turnover (Water consumption / Revenue from operations)	4.48×10^{-5}	5.13×10^{-5}
Water intensity per rupee of turnover adjusted for Purchasing Power Parity (PPP) (Total water consumption / Revenue from operations adjusted for PPP)	0.90×10^{-3}	1.17×10^{-3}
Water intensity in terms of physical output	-	-
Water intensity (optional) - the relevant metric may be selected by the entity	-	-

Representation of Water intensity values by NFIL in exponential form, also use of hyphen for nil values in FY 2024-25 report

Essential Indicator 4: Has the entity implemented a mechanism for Zero Liquid Discharge?
If yes, provide details of its coverage and implementation.

Total water discharged and assurance for the last three financial years

Company	Financial Year	Total Water discharged	Assurer for ZLD
Tata Chemicals Ltd (TCL)	2022 - 23 2023 - 24 2024 - 25	85,017,300 85,022,898 81,827,000	Ernst & Young LLP (EY) KPMG Assurance and Consulting Services LLP KPMG Assurance and Consulting Services LLP
Gujarat Fluorochemicals Ltd (GFL)	2022 - 23 2023 - 24 2024 - 25	971,214 1,007,020 1,123,530	Gujarat Pollution Control Board (GPCB) & Ernst & Young LLP (EY) Intertek India Private Limited Intertek India Private Limited
Navin Fluorine International Ltd (NFIL)	2022 - 23 2023 - 24 2024 - 25	469,605 558,766 555,597	False (As per BRSR report) False (As per BRSR report) TÜV SÜD South Asia Pvt. Ltd

Zero Liquid Discharge disclosure

Company	ZLD implementation
Tata Chemicals Ltd (TCL)	The Company has implemented zero liquid discharge mechanism at Mambattu plant, Akola plant, and Wyoming plant. Ankleshwar and Dahej CZ plant developed the capability for 100% recycling of treated water.
Gujarat Fluorochemicals Ltd (GFL)	Yes, Ranjitnagar unit has implemented the Zero Liquid Discharge (ZLD). The unit has installed a Multi Effect Evaporator (MEE) and Reverse Osmosis (RO) system as part of the tertiary treatment, and all the water is being recycled and used for plantation activity within the premises.
Navin Fluorine International Ltd (NFIL)	Yes, the Company's wastewater treatment plant at the Dewas site operates on a Zero Liquid Discharge (ZLD) principle, incorporating an Effluent Treatment Plant, Reverse Osmosis, and Multi Effect Evaporator Plant. The treated effluent is utilized for various purposes, including within its utility systems such as the cooling tower. Similarly, at the Company's subsidiary location in Dahej, a wastewater treatment facility designed to address specific characteristics through primary, biological, and tertiary treatment methods has been implemented. To manage concentrated streams, a Multi Effect Evaporator is utilized, ensuring compliance with permissible limits before discharge to the industrial estate wastewater point. Meanwhile, in Surat, the manufacturing site treats wastewater through a three-stage process before discharging it to the common treatment plant. The treated water is then recycled onsite to support utility needs.

Break-up of water discharged by destination & treatment for 2024-2025

Discharge	Tata Chemicals Ltd (TCL)	Gujarat Fluorochemicals Ltd (GFL)	Navin Fluorine International Ltd (NFIL)
To Surface Water	14,182,400	0	0
To Groundwater	0	0	0
To Seawater	64,351,231	883,092	283,243
By Sent To Third Parties	486,072	240,436	272,354
To Others	2,807,305	0	0
Total Water Discharged In Kilolitres	81,827,012	1,123,528	555,597

Water Intensity peer comparison among chemical industries

Company	Financial Year	Total Water discharged	Assurer for ZLD
Tata Chemicals Ltd (TCL)	2022 - 23 2023 - 24 2024 - 25	85,017,300 85,022,898 81,827,000	Ernst & Young LLP (EY) KPMG Assurance and Consulting Services LLP KPMG Assurance and Consulting Services LLP
Gujarat Fluorochemicals Ltd (GFL)	2022 - 23 2023 - 24 2024 - 25	971,214 1,007,020 1,123,530	Gujarat Pollution Control Board (GPCB) & Ernst & Young LLP (EY) Intertek India Private Limited Intertek India Private Limited
Navin Fluorine International Ltd (NFIL)	2022 - 23 2023 - 24 2024 - 25	469,605 558,766 555,597	False (As per BRSR report) False (As per BRSR report) TÜV SÜD South Asia Pvt. Ltd

Independent Assurance Status

Company	Independent Assurance Provider	Scope of Assurance	Level / Nature of Assurance	Coverage
Tata Chemicals Ltd (TCL)	KPMG Assurance & Consulting Services LLP	BRSR Core indicators	Reasonable Assurance	Explicitly covered, including water withdrawal & discharge.
Gujarat Fluorochemicals Ltd (GFL)	Intertek India Pvt. Ltd	BRSR Core indicators	Reasonable Assurance	Gujarat Pollution Control Board (GPCB) as independent auditor.
Navin Fluorine International Ltd (NFIL)	TÜV SÜD South Asia Pvt. Ltd	BRSR Core indicators	Reasonable Assurance	Explicitly covered, including water withdrawal & discharge.

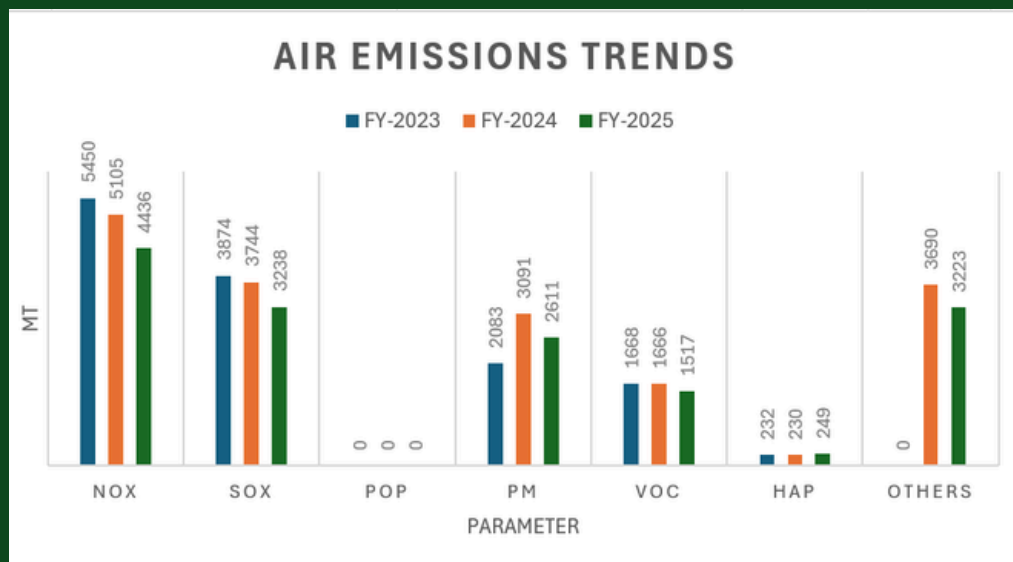
Essential Indicator 6 : Details of air emissions (other than GHG emissions) by the entity.

Tata Chemicals Limited has consistently reported their NOx (nitrogen oxides), SOx (sulphur oxides), PM (particulate matter), VOC (volatile organic compounds), HAP (hazardous air pollutants), and CO (carbon monoxide) emissions for the Financial Years 2022-23, 23-24, and 24-25 under the 'air emissions (other than GHG)' section (Sec C, P6, EI6) of the BRSR report. The unit they have used is Metric Tonnes for all these parameters.

Year on Year trends of Tata Chemicals Limited

The noticeable data gap was in the POP (persistent organic pollutants) and Others (please specify) disclosure, where in FY 2022-23 it was shown as 0 - it is not clear whether the real amount of POP was 0 or simply not measured, since in this year, while they have received assurance from EY, it is not mentioned whether the assurance was limited or reasonable. This is a glaring omission of information and makes the reported figure of 0 in this year doubtful for these 2 parameters.

In 2023-24, the levels are 0 for POP, assured by KPMG, but again not mentioned if it is limited or reasonable. CO was mentioned under 'Others'. In 2024-25, POP was given as Not Applicable, but in the previous 2 years it was reported as 0. No reasoning has been given for this sudden shift.



In 2024-25, CO again mentioned under 'Others', with a different number from the previous year, showing that it was not simply copy-pasted. However, even in 2024-25, it is still not mentioned whether the KPMG assurance was limited or reasonable.

6. Please provide details of air emissions (other than GHG emissions) by the entity, in the following format:

Parameter	Please specify unit	FY 2024-25	FY 2023-24
NOx	MT	4,436	5,105
SOx	MT	3,238	3,744
Particulate matter (PM)	MT	2,611	3,107
Persistent organic pollutants matter (POP)	MT	NA	NA
Volatile organic compounds (VOC)	MT	1,517	1,666
Hazardous air pollutants (HAP)	MT	249	230
Others - Carbon Monoxide (CO)	MT	3,223	3,690

Note: Indicate if any independent assessment/evaluation/assurance has been carried out by an external agency? (Y/N) If yes, name of the external agency.

Assurance has been carried out by an external agency? (Yes / No) Yes
Name of external agency KPMG Assurance and Consulting Services LLP

Tata Chemicals Limited has been highly transparent & explicit in their PM reporting, wherein in 2023-24, they have done granular reporting of PM in 2.5 microns and 5 microns (μm) & clearly stated that they have revised the figures of 2022-23 while restating in 2023-24, to match the new methodology used in 2023-24 and ensure comparability & continuity of information across these 2 years. They have admitted to a marked increase in their 2022-23 emissions of over 80% as per the new methodology. This shows excellent governance oversight & true ESG leadership on the part of this company.

Please provide details of air emissions (other than GHG emissions) by the entity in the following format:

Parameter	Please specify unit	FY 2024-25 (Current Financial Year)	FY 2023-24 (Previous Financial Year)
NOx	MT/Year	98	86
SOx	MT/Year	86	162
Particulate Matter (PM)	MT/Year	74	40
Persistent Organic Pollutants (POP)	NA	NA	NA
Volatile Organic Compounds (VOC)	NA	NA	NA
Hazardous Air Pollutants (HAP)	NA	NA	NA
Others - please specify	NA	NA	NA

Note: Indicate if any independent assessment/evaluation/assurance has been carried out by an external agency? (Y/N) If yes, the name of the external agency.

Gujarat Fluorochemicals Limited reported emissions of NOx, SOx, and PM, but left the fields for POPs, VOCs, HAPs, and others as "NA." This shows that the company has not thoroughly assessed the impact of their operations on these parameters & needs more rigorous governance to spearhead reporting. These parameters are highly relevant to a chemicals company, hence they need to be more diligent instead of simply marking these as "NA". Limited assurance undertaken by Intertek India Private Limited.

6. Please provide details of air emissions (other than GHG emissions) by the entity, in the following format:

Parameter	Unit	FY 2024-25	FY 2023-24
NOx	MT	18.76	11.42
SOx	MT	15.87	11.37
Particulate matter (PM)	MT	9.65	9.67
Persistent organic pollutants (POP)	-	Nil	Nil
Volatile organic compounds (VOC)	-	Nil	Nil
Hazardous air pollutants (HAP)	-	Nil	Nil
Others - please specify	-	Nil	Nil

Navin Fluorochemicals Limited has only reported data for NOx, SOx, and PM; "Nil" has been entered in the template for the remaining parameters (POP, VOC, HAP, others), but the units are missing. It is unclear whether assessments were done for these parameters & came out to be 0, or whether they were not done at all. No mention of assurance for this indicator.

Essential Indicator 7: Details of greenhouse gas emissions (Scope 1 and Scope 2 emissions) & their intensity.

Tata Chemicals Limited has demonstrated consistent transparency and accuracy in its GHG emissions reporting framework from FY 2022-23 to FY 2024-25. The company has reported Scope 1 and Scope 2 emissions in metric tons of CO₂ equivalent. External assurance has been provided by EY for 2022-23 and by KPMG for 2023-24, 2024-25. The company has ensured that they have restated data for FY 2023-24 in FY 2024-25 to ensure comparability, reflecting their commitment to clean and consistent reporting. They have also specified the exact percentages of difference in the emissions levels which emerged due to the change of methodology. Overall, the trend shows that the company's total emissions have increased since 2022, & the emissions intensity by rupee of revenue too has increased. Hence, they need to ramp up their decarbonisation efforts to match the efforts they have put into reporting transparency.

Year on Year trends of Tata Chemicals Limited Absolute numbers of Scope 1 and Scope 2

Parameter	Unit	2024-25	2023-24	2022-23
Total Scope 1	MT of CO2 eq	4,770,357	4,424,759	4,363,737
Total Scope 2	MT of CO2 eq	75,534	69,786	77,772

Intensity figures of Scope 1 and Scope 2 (not adjusted for Purchasing Power Parity)

Parameter	Parameter	2024-25	2023-24	2022-23
Total Scope 1 & Scope 2 emissions intensity per rupee of turnover	MT of CO2 eq/INR	0.000033	0.000029	0.000026

Peer Comparison for 2024-25

Parameter	Unit	Tata Chemicals	Gujarat Fluorochemicals	Naveen Fluorine International
Total Scope 1	MT of CO2 eq	4,770,357	709,492	164,451.34
Total Scope 2	MT of CO2 eq	75,534	402,888	46,196.79
Total Scope 1 & Scope 2 intensity / rupee turnover	MT of CO2 eq/INR	0.000033	0.000024	8.97×10^{-5}

It can be observed that Gujarat Fluorochemicals has the highest Scope 2 emissions amongst all, while Naveen seems to have the highest emissions intensity among all. Being a smaller company than the rest, Naveen needs to invest more in improving its operational efficiency in anticipation of a future where its business expands and it will inevitably face more pressure to reduce emissions per rupee of turnover. It seems that both Gujarat and Naveen can take notes from Tata in terms of minimizing their Scope 2 emissions compared to the size of their operations.

While Tata Chemicals has been diligent in revising their 2023-24 numbers, they have not done the same for 2022-23 in accordance with their policy of providing continuity of information. Hence, we are left with uncertainty as to whether the numbers for Scope 1 and Scope 2 for that year are accurate or not.

Essential Indicator 9: Details related to waste management by the entity.

During the analysis of the BRSR reports submitted by the Tata Chemicals Limited for FY2022-23, FY2023-24, and FY2024-25, the disclosures provided was found to have few mismatches. The reports did align to the required formats of BRSR. All of the provided values associated with the amount of wastes generated, disposed and recovered are provided in MT, that aligns with the given BRSR criteria.

The waste generated has reduced prominently after FY2022-23. However, for every year most of the wastes are being contributed by the “other hazardous wastes” category, except for the Tata Chemicals Limited, where most of the wastes are being contributed by the “Other non-hazardous waste”

Year on Year trends of Tata Chemicals Limited

During the analysis of the BRSR reports submitted by the Tata Chemicals Limited for FY2022-23, FY2023-24, and FY2024-25, the disclosures of values provided for other hazardous waste provided for FY2023-24 for the reports of 2024 and 2025 were found to be different. The same mismatched errors were also observed other non-hazardous waste provided for FY2023-24 for the reports of 2024 and 2025. These types of errors in the report has made the disclosures by the Tata Chemicals Limited unreliable for the readers.

Waste Intensity

Year on Year trends of Tata Chemicals Limited: The “Waste Intensity” was reported for only FY2023-24 and FY2-24-25, and not for the FY2022-23 in the BRSR reports. The “Waste Intensity per Turnover” remained constant for the past two years’ reports of BRSR.

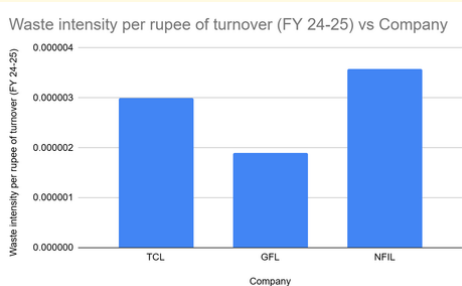


Fig. Waste intensity per rupee of turnover (FY 2024-25)

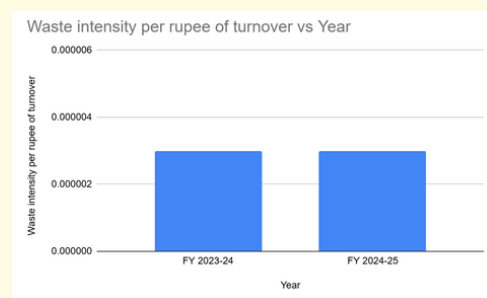


Fig. Waste intensity per rupee of turnover (TATA)

Peer Comparison of Tata Chemicals Limited with Gujarat Fluorochemicals Limited and Navin Fluorine International Limited : Waste Intensity per Turnover reported by the three companies for the FY2024-25, shows that Navin Fluorine International Limited has the highest value for Waste Intensity per turnover, despite of the lowest waste generation in comparison to the other two companies, that indicates that the waste management is highly inefficient for NFIL and also the lowest turnover generated by the company.

Peer Comparison of Tata Chemicals Limited with Gujarat Fluorochemicals Limited and Navin Fluorine International Limited

Navin Fluorine International Limited: The total waste generated has increased in 2024, possibly due to the increase in the wastes under other hazardous wastes. The waste intensity in terms of physical output has also not been reported in the report.

Gujarat Fluorochemicals Limited: According to the BRSR PDF- Total waste generated (89,447MT), which is not equal to wastes recovered (85,884.3MT)+ wastes disposed (3563MT) = 89447.3MT. Also, for the waste disposed does not actually add up to the total that they have mentioned. Mismatches were also found in the disclosed values when the XBRL type was compared to the PDF type for the “Total waste generated” and the “Total waste disposed.”

Tata Chemicals Limited: The generation of wastes like E-waste, Bio-medical waste and battery waste have increased prominently. The disclosures provided are almost reliable. However, a mismatched disclosure of value for the total generation of wastes was found while comparing the XBRL report with that of the PDF type.

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All the XBRL Files were downloaded from the NSE Website : <https://www.nseindia.com/companies-listing/corporate-filings-bussiness-sustainability-reports>